

DTT DIV in-vessel coil load model for disruption studies

Simscape (.ssc) implementation, identification on the ENEA specification curves and validation

Technical note – Rev. 00 – 2026-09-18

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Revision history

Rev.	Date	Description
00	2026-09-18	First issue. Load model of the DIV coils with the plasma-induced e.m.f., identified on Table 2, Fig. 3, Fig. 6 and Fig. 7 of ENEA specification PSS-SPT-59401 rev. 1.0; Simscape components <code>dttdivcoil</code> , <code>dttdivcoils3</code> , <code>dttemf_table</code> , <code>dttemf_analytic</code> , <code>dttplasma_emf</code> ; MATLAB parameter and validation scripts; digitised curves. Companion of the crowbar note [2] and of the L_c reactor note [3].

Abstract

This note delivers a Simscape model of the DTT divertor in-vessel coils (DIV1, DIV2, DIV3) as seen from the power-supply terminals during a plasma disruption, so that the complete converter – filter – crowbar – protection-coil circuit can be simulated in Simulink and the feasibility of the protection strategies can be documented. The coil is a Thevenin equivalent: the open-circuit e.m.f. $e(t)$ induced by the plasma (specification Fig. 6, digitised) in series with the coil resistance, self-inductance and two eddy-current loops identified on the frequency-dependent impedance of specification Fig. 3. The equations are exact for a linear system after the current quench and are implemented in Simscape language with the sign convention of the specification. The identification shows a fact that must be known before using the specification curves: the Fig. 7 crowbar current (10.9 kA peak, 4.8 kA after 300 ms) is *not* the response of the specified circuit to the Fig. 6 e.m.f., which gives only 6.0 kA; it is reproduced within 0.24 kA rms by the circuit of specification §8.2 ($L_{coil} + L_{cab} + L_p = 3.46$ mH, $R_{coil} = 24.9$ m Ω) driven by 1.84 times the Fig. 6 e.m.f. Both worst cases are therefore provided as parameter sets (*voltage* worst case $k_e = 1$, *current* worst case $k_e = 1.84$), together with a black-box R-L equivalent and a parametrised plasma-flux model (VDE growth, linear current quench, passive-structure delay) for what-if studies. The influence of the elements neglected by ENEA (cable and protection-coil resistance, eddy currents), of the quench duration and of the crowbar closing time is quantified, and the questions to be put to ENEA are listed.

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1 Scope and summary

The crowbar note [2] established the duty of the DIV crowbar from the worst-case curves of the ENEA specification [1] and the reactor note [3] designed the series reactor L_c of the diode-bridge crowbar imposed by the customer. Both used a time-domain replica of the specification curves and simple R-L circuits. The next step, requested by the customer, is a load model usable inside a complete Simulink/Simscape model of the power supply (12-pulse rectifier, DC link, H-bridge, output filter, crowbar, protection coil), so that every situation – disruption with the converter in current control, with the crowbar closed at different instants, with the fast dump active, with the converter in “protection mode”, with the coils coupled – can be tried and the limits of the decisions taken in the past can be documented with numbers.

The model is a Thevenin equivalent of the coil (Sec. 3): an e.m.f. $e(t)$ in series with the coil resistance R_0 , the self-inductance L_0 and two eddy-current loops that represent the vacuum vessel and the passive structures. The e.m.f. is the open-circuit voltage induced by the plasma, i.e. specification Fig. 6, and the eddy loops are identified on the frequency-dependent inductance and resistance of specification Fig. 3. Five Simscape components are delivered (Sec. 5): the single coil, the three coupled coils, a table-driven e.m.f. source loaded with the digitised Fig. 6 curves, an analytic e.m.f. source parametrised by flux, growth rate, quench time and passive-structure delay, and an e.m.f. source driven by externally supplied plasma current and plasma-coil mutual inductance.

The identification (Sec. 4) produced one result that changes how the specification curves must be read. Fig. 6 and Fig. 7 of [1] are not consistent with each other through the circuit that the specification itself defines in §8.2: with $L = L_{coil} + L_{cab} + L_p = 3.46$ mH and $R = R_{coil} = 24.9$ m Ω (§8.4 states that the cable and protection-coil resistances were not considered in Fig. 7) the Fig. 6 e.m.f. produces a crowbar current of 6.0 kA, not 10.9 kA. The Fig. 7 curve is reproduced within 0.24 kA rms – pre-quench dip, peak, time of the peak and tail – by the same circuit driven by an e.m.f. 1.84 times larger than Fig. 6. Fig. 6 and Fig. 7 are therefore worst cases of two different quantities (voltage and current), presumably from two different disruption cases of the ENEA database, and a model that reproduces both needs a gain k_e on the e.m.f.: $k_e = 1$ for the open-circuit voltage and the converter overvoltage, $k_e = 1.84$ for the crowbar current and I^2t . A single R-L black box (1.92 mH, 13.1 m Ω , [2]) reproduces Fig. 7 equally well but with an inductance and a resistance that

have no physical meaning and that would falsify any simulation in which the converter is active; it is kept only as a cross-check.

The consequences for the protection studies are summarised in Sec. 4 and Sec. 6: the induced current is imposed by the flux swing and not by the voltage peak (the peak current changes by less than 1% when the quench duration is varied from 1.5 ms to 12 ms, while the voltage peak changes from 6.5 kV to 2.5 kV); 92% of that flux reaches the coil through the passive structures with a 26 ms time constant, so the crowbar closing instant within a few milliseconds of the quench is irrelevant for the peak; the elements neglected by ENEA reduce the peak by 3–5 % and the I^2t by 16–18 %; the protection coil halves the current that the crowbar would see without it. The main limits of the model are that the plasma response before the quench and the coil–coil mutual inductances are not available from the specification (Sec. 6).

2 Reference data

2.1 Stand-alone parameters and frequency-dependent impedance

Table 2 of [1] gives the stand-alone coil parameters, the worst-case connection parameters (cables) and the protection-coil inductances (Table 1). Fig. 3 of [1] gives the equivalent inductance and resistance of the “complete circuits” (load coil plus protection coil, without connections) between 0 Hz and 10 Hz, computed by ENEA with the coupling to the passive structures. The inductance falls and the resistance rises with frequency (Fig. 1); the DC values (1.92, 1.68, 3.12 mH) are 4–7 % lower than the stand-alone sums (2.0, 1.8, 3.3 mH), which the specification attributes to “coupling effects”. Note that the specification defines the Fig. 3 curves as “the minimum achievable in the circuits” and requires the Contractor to consider them.

Table 1: Load data from [1], Table 2 and Fig. 3.

	DIV1	DIV2	DIV3
Stand-alone inductance L_{coil} [mH]	0.5	1.2	1.7
Stand-alone resistance R_{coil} [mΩ]	10.6	18.7	24.9
Protection coil L_p [mH] ($\pm 5\%$)	1.5	0.6	1.6
Connection (worst case, cables)	160 μH, 5 mΩ		
Fig. 3 equivalent L at DC / 10 Hz [mH]	1.92 / 1.84	1.68 / 1.46	3.12 / 2.77
Fig. 3 equivalent R at DC / 10 Hz [mΩ]	10.6 / 15.5	18.7 / 33.0	24.9 / 44.1
Open-circuit e.m.f. peak, Fig. 6 [kV]	−5.9	−5.9	−2.9
Crowbar current, Fig. 7 [kA]	–	–	10.9 (+5 = 15.9)

2.2 Disruption curves

Fig. 6 of [1] gives the voltage induced in the coils in the worst-case disruption with the coil circuit open; Fig. 7 gives, for DIV3 and an ideal crowbar, the induced overcurrent and its sum with the 5 kA pre-disruption current. Both were digitised in [2] at 0.05 ms steps over 0–300 ms; the curves are shown in Fig. 2 and are delivered as `data/dtt_fig6_fig7_digitised.csv` after a light median filter that removes the grid-line artefacts of the digitisation without touching the quench spike (peak and flux preserved within 0.1%). The accuracy is about 1% of full scale.

The DIV3 e.m.f. has three phases: a slow positive growth to 344 V between 0 and 80 ms (the plasma drifts towards the coil during the vertical displacement event, VDE), a negative spike to −2.86 kV at 83.3 ms about 3 ms wide (the current quench), and a negative tail that decays with a time constant of about 25 ms (the currents induced in the vessel by the quench decay and their flux leaves the coil). The flux linked by the coil, $\int e dt$, is 7.2 Wb in the first phase, −36.8 Wb in the spike and tail, −29.6 Wb in total; for DIV1/DIV2 the total is −60.6 Wb. The tail carries most of the flux: only about 8% of the quench flux reaches the coil during the 3 ms spike, the rest arrives through the passive structures over the following 50–100 ms.

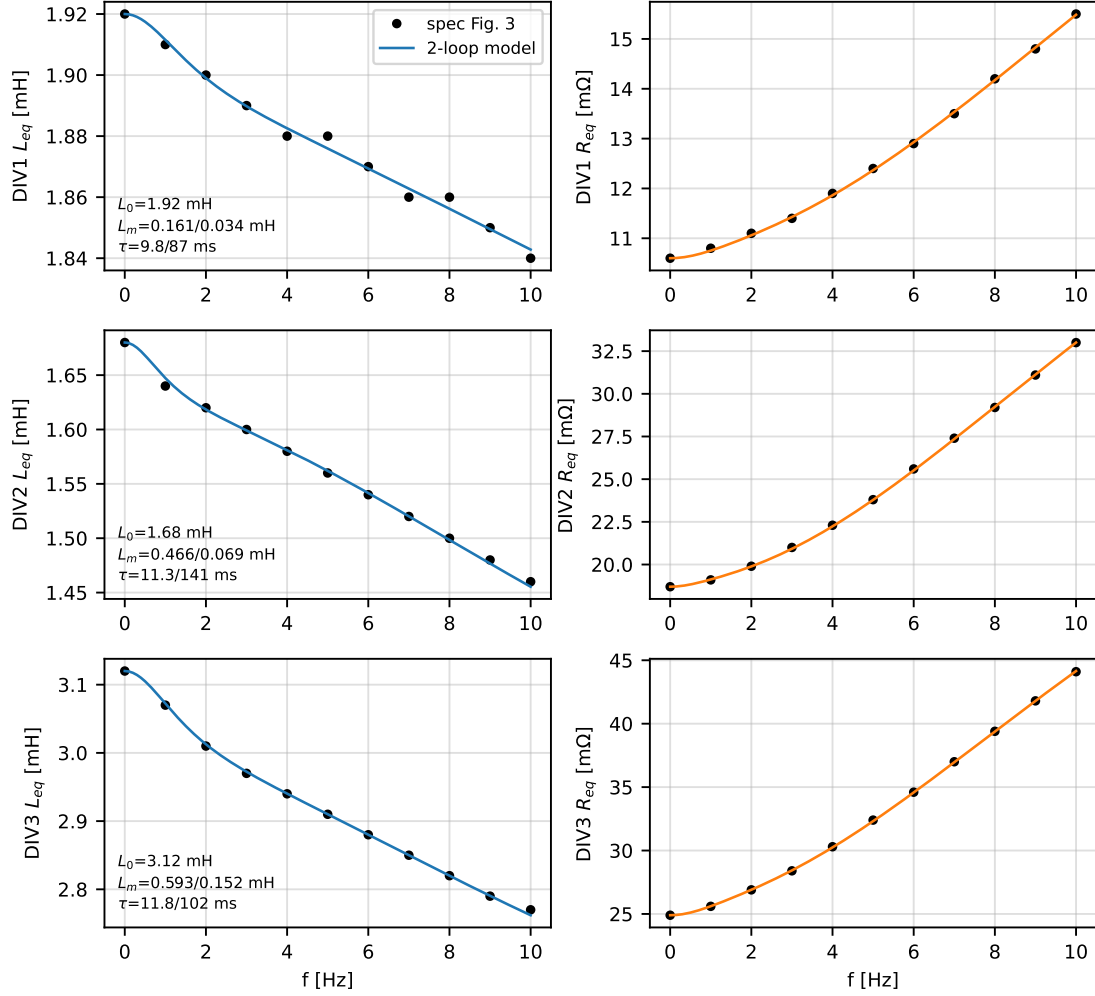


Figure 1: Equivalent inductance and resistance of the three DIV circuits (coil + protection coil) from [1] Fig. 3 (dots, values read from the figure labels) and the two-loop eddy-current model of Sec. 3.2 (lines).

The Fig. 7 induced current starts negative (-3.2 kA at 80 ms, minimum -3.4 kA, in response to the positive pre-quench e.m.f.), crosses zero 2 ms after the spike, peaks at 10.9 kA at 139 ms, i.e. 56 ms after the spike, and is 4.8 kA at 300 ms. Its I^2t over the 300 ms window is 14.6 MA²s (38.1 MA²s for the total current with the constant 5 kA, 44 MA²s with the tail extrapolated to zero as done in [2]).

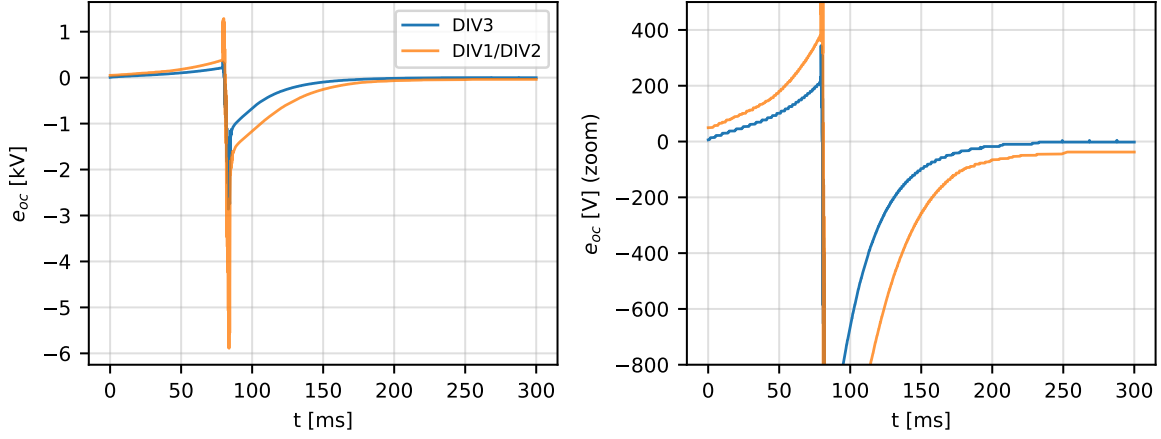


Figure 2: Digitised open-circuit e.m.f. of [1] Fig. 6 (ENEA sign convention): full scale (left) and zoom on the pre-quench growth and on the tail (right). These curves are the input $\mathbf{e}$ of the load model.

2.3 Sign convention

The specification plots the induced voltage positive during the VDE and negative at the quench, and the crowbar current positive. The model keeps exactly this convention: with the coil terminal p connected to the positive output of the converter and the current i counted into p , the terminal equation is $v = R_0 i + L_0 di/dt + \dots + k_e e(t)$ with $e(t)$ the Fig. 6 curve as plotted. A short circuit ($v = 0$) then gives $i = -k_e e/(R + sL)$, i.e. the positive current of Fig. 7. The worst case of §8.4 (pre-disruption current “in the same direction of the induced overcurrent”) is $i_0 = +5$ kA; the opposite polarity of the disruption is obtained with $k_e < 0$, as §8.5 requires the crowbar to be bidirectional because “the curves in Figure 7 can occur in both directions”.

3 Model equations

3.1 Thevenin equivalent with eddy-current loops

As long as the plasma current is an imposed source and the vessel, the passive structures and the other coils are linear, superposition applies to the coil terminals: the terminal voltage is the open-circuit response to the plasma (which contains the response of the passive structures to the plasma) plus the response to the coil current with the plasma removed,

$$v(s) = e_{oc}(s) + Z(s) i(s), \quad Z(s) = R_0 + sL_0 - \sum_k \frac{s^2 M_k^2}{R_k + sL_k}, \quad (1)$$

where $Z(s)$ is the driving-point impedance of the coil coupled to N passive loops (L_k , R_k , mutual M_k). This is exact after the current quench, when there is no plasma left and the system is passive; before the quench the plasma responds to the coil current (vertical instability) and Eq. (1) neglects that response, which the specification does not provide either. Only the two combinations $L_{m,k} = M_k^2/L_k$ and $\tau_k = L_k/R_k$ enter $Z(s)$, so each loop is characterised by two numbers and can be referred to the coil by choosing $M_k = L_k = L_{m,k}$; its current x_k then has the scale of the coil

current. In the time domain (Fig. 3)

$$v = R_0 i + L_0 \frac{di}{dt} + \sum_{k=1}^N L_{m,k} \frac{dx_k}{dt} + k_e e(t), \quad (2)$$

$$0 = \tau_k \frac{dx_k}{dt} + x_k + \tau_k \frac{di}{dt}, \quad k = 1 \dots N. \quad (3)$$

The frequency-dependent inductance and resistance seen at the terminals are

$$L(\omega) = L_0 - \sum_k L_{m,k} \frac{(\omega\tau_k)^2}{1 + (\omega\tau_k)^2}, \quad R(\omega) = R_0 + \sum_k L_{m,k} \frac{\omega^2\tau_k}{1 + (\omega\tau_k)^2}, \quad (4)$$

which is the form of the specification Fig. 3: L drops from L_0 towards $L_0 - \sum_k L_{m,k}$ and R rises from R_0 , each loop contributing around $\omega \approx 1/\tau_k$. Eqs. (2)–(3) are what the Simscape component `dttdiv_coil` solves, with $N = 2$; the cable and the protection coil are left outside the component because they belong to the power supply and the user will want to vary them.

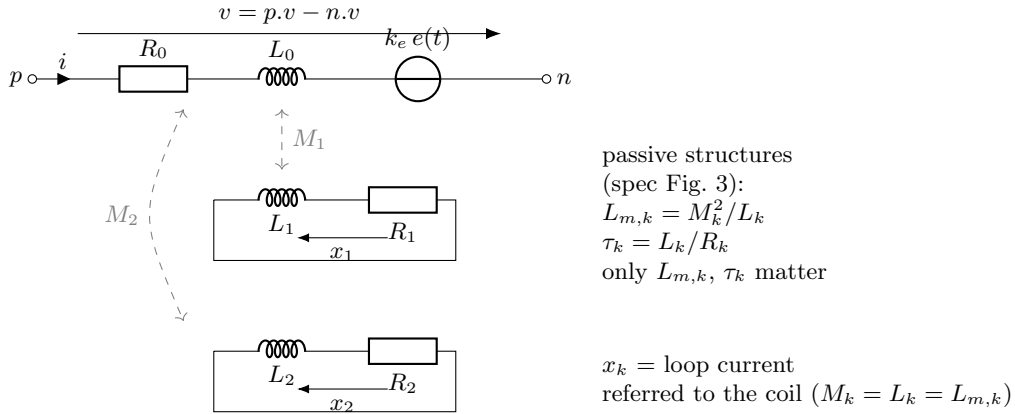


Figure 3: Equivalent circuit implemented in `dttdiv_coil`: coil branch with the e.m.f. and two eddy-current loops coupled to it.

3.2 Identification of the eddy-current loops on Fig. 3

With L_0 and R_0 fixed at the DC values of Fig. 3, the loop parameters were fitted by least squares on the eleven labelled points of each curve (Table 2, Fig. 1). One loop is not enough (residual 41 μH and 0.66 $\text{m}\Omega$ on DIV3); two loops reproduce all six curves within the resolution of the figure labels (3 μH , 0.05 $\text{m}\Omega$); a third loop is not constrained by data limited to 10 Hz. The time constants are similar for the three coils (10–12 ms and 90–140 ms), as expected for structures common to the three coils, and the coupling is strongest for DIV2 ($\sum L_m/L_0 = 32\%$) and DIV3 (24%). The extrapolation of Eq. (4) above 10 Hz is a hypothesis of the model: the high-frequency inductance $L_0 - \sum L_m$ (2.38 mH for the DIV3 circuit with L_p) is what the coil shows to the 3 ms quench spike, and faster loops that Fig. 3 cannot reveal would lower it further. Because 92% of the disruption flux reaches the coil on the 25 ms time scale, this uncertainty has little effect on the crowbar current (Sec. 4.2) but it matters for the voltage seen by the converter during the spike.

3.3 Analytic plasma-flux model of the e.m.f.

For what-if studies (different quench durations, larger flux, disruption at a different instant, opposite polarity) a parametrised e.m.f. is more useful than a table. The physical picture of [1] §8.2 – the coil links a flux $\Psi = M_{cp}I_p$ with the plasma – is represented as follows. During the flat top the flux $\Psi_a = M_{cp,0}I_{p0}$ is constant and induces nothing; during the VDE the plasma approaches the coil, M_{cp} grows and the flux increases by $\Delta\Psi_v$ with growth rate g ; at t_q the plasma current is quenched

Table 2: Two-loop eddy-current model of the three DIV circuits (coil + L_p) fitted on [1] Fig. 3. The loops are attributed to the coil (the air-core protection coil has no frequency dependence); L_0 of the coil alone is $L_{dc} - L_p$ or the stand-alone value of Table 2.

Circuit	L_{dc} [mH]	R_0 [mΩ]	$L_{m,1}$ [mH]	τ_1 [ms]	$L_{m,2}$ [mH]	τ_2 [ms]	$L_{dc} - \sum L_m$ [mH]	rms L / R
DIV1	1.92	10.6	0.161	9.8	0.034	86.9	1.73	2 μH / 0.03 mΩ
DIV2	1.68	18.7	0.466	11.3	0.069	140.9	1.15	3 μH / 0.05 mΩ
DIV3	3.12	24.9	0.593	11.8	0.152	102.3	2.38	3 μH / 0.04 mΩ

linearly to zero in t_{cq} ; the vessel and the passive structures delay a fraction γ of the flux variation with a time constant τ_v :

$$\Psi(t) = \begin{cases} \Psi_a & t \leq 0 \\ \Psi_a + \Delta\Psi_v \frac{e^{g(t-t_q)} - e^{-gt_q}}{1 - e^{-gt_q}} & 0 < t \leq t_q \\ (\Psi_a + \Delta\Psi_v) \left(1 - \frac{t-t_q}{t_{cq}}\right) & t_q < t < t_q + t_{cq} \\ 0 & t \geq t_q + t_{cq} \end{cases} \quad (5)$$

$$e(t) = (1 - \gamma) \frac{d\Psi}{dt} + \gamma x, \quad \tau_v \frac{dx}{dt} + x = \frac{d\Psi}{dt}. \quad (6)$$

The seven parameters were fitted on the digitised Fig. 6 (Table 3, Fig. 4). For DIV3 the fit reproduces the pre-quench growth, the tail and the total flux (-30.1 vs -29.6 Wb) within 53 V rms; the single-time-constant delay smooths the spike (-2.30 vs -2.86 kV), so for the converter overvoltage the digitised table is to be preferred, whereas for the crowbar current the analytic source gives the same result as the table within 2% (11.28 vs 11.05 kA in the circuit of Sec. 4). The fitted values have a plausible physical reading: with $I_{p0} = 5.5$ MA the plasma-coil mutual inductance is 5.6 μH at flat top and 7.7 μH at the quench for DIV3, the current quench lasts 3.2 ms ($dI_p/dt \approx 1.7$ GA/s, in the range of the ITPA database [10] for a machine of this size), and 92% of the flux reaches the coil through the vessel with $\tau_v = 26$ ms, which is the order of magnitude of the DTT vacuum-vessel time constant. The DIV1/DIV2 fit is poorer (116 V rms) because their pre-quench curve has a sharp positive spike just before the quench (thermal quench) that Eq. (5) does not describe.

Table 3: Analytic e.m.f. model fitted on [1] Fig. 6 (defaults of `dtb.emf_analytic` and `par.plasma` in `dtb_div_params.m`).

Coil	Ψ_a [Wb]	$\Delta\Psi_v$ [Wb]	g [1/s]	t_q [ms]	t_{cq} [ms]	τ_v [ms]	γ	rms [V]
DIV3	30.6	11.9	31.8	81.1	3.19	25.8	0.92	53
DIV1/DIV2	62.9	24.7	1.3	81.6	2.70	37.6	0.91	116

3.4 Three coupled coils

A disruption induces an e.m.f. in the three coils at the same time (Fig. 6), all three crowbars may close and the induced currents are coupled through the mutual inductances M_{12} , M_{13} , M_{23} and through the common passive structures. The component `dtb.div_coils3` implements

$$v_j = R_{0,j} i_j + \sum_{l=1}^3 L_{jl} \frac{di_l}{dt} + \sum_{m=1}^2 L_{m,jm} \frac{dx_{jm}}{dt} + k_{e,j} e_j(t), \quad \tau_{jm} \frac{dx_{jm}}{dt} + x_{jm} + \tau_{jm} \frac{dj}{dt} = 0, \quad (7)$$

with the 3×3 inductance matrix $\mathbf{L}$ as a parameter. The specification does not give the mutual inductances (the default is $M = 0$, i.e. three independent coils); they must be requested from ENEA, together with the plasma-induced e.m.f. of the three coils for the same disruption case. An attempt to infer the mutuals from Fig. 7 alone (three coils shorted, Fig. 6 e.m.f.) was made and

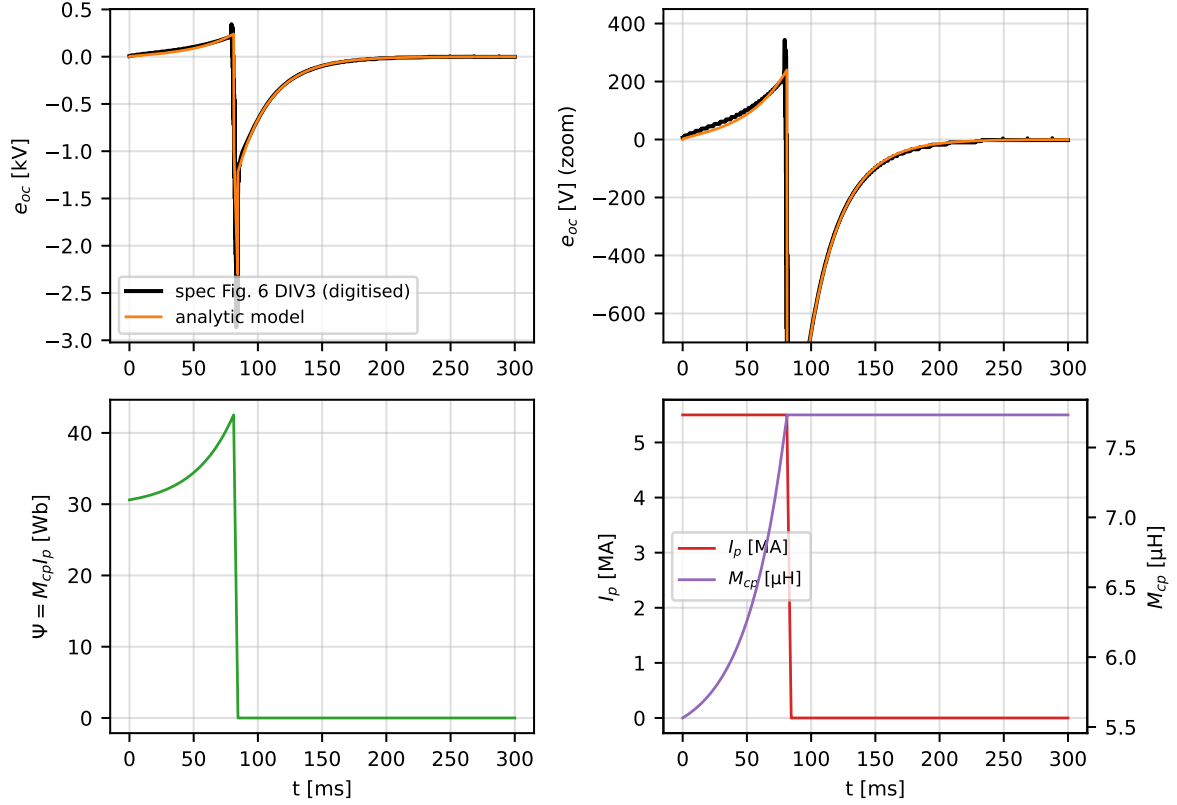


Figure 4: Analytic e.m.f. model (Eqs. (5)–(6)) against the digitised Fig. 6 for DIV3 (top), and the underlying flux, plasma current and mutual inductance trajectories (bottom) used by `dtf.emf_analytic` and generated by `dtf.plasma_scenario.m` for `dtf.plasma_emf`.

discarded: it requires negative mutuels of 0.4–0.7 mH and still misses the tail of Fig. 7, so the Fig. 7 current is not explained by coil–coil coupling.

4 Reproduction of the specification curves

4.1 Fig. 6 does not produce Fig. 7 through the specified circuit

The specification states in §8.2 that the overcurrent with an ideal crowbar is given by $v_{induced} \simeq (L_{coil} + L_{cable} + L_p) di/dt + (R_{coil} + R_{cable} + R_p) i$ and in §8.4 that “the actual overcurrent will be lower due to the effect of the resistance of the protection coil and of the connections (both not considered in the simulation in Figure 7)”. The circuit of Fig. 7 is therefore $L = 1.7 + 0.16 + 1.6 = 3.46$ mH (or 3.30 mH without the cable) and $R = 24.9$ m Ω , $\tau = L/R = 139$ ms. Driving this circuit with the digitised Fig. 6 DIV3 e.m.f. gives a peak of 6.0 kA at 139 ms and 2.35 kA at 300 ms (Fig. 5, green): the shape of Fig. 7 but 55% of its amplitude. The eddy loops of Fig. 3 change this by less than 3% (6.7 kA), a hypothetical fast loop tuned on the spike does not help because the flux arrives slowly, and a resistance free to vary reaches at most 8.2 kA with 7 m Ω while spoiling the tail. In [2] Fig. 7 was reproduced with a free R-L fit, $L = 1.92$ mH and $R = 13.1$ m Ω (rms 0.21 kA); both values are 1.8–1.9 times smaller than those of the specified circuit while their ratio, the time constant, is the same (146 vs 139 ms).

This last observation gives the consistent explanation: since the circuit is linear, dividing L and R by the same factor is the same as multiplying the e.m.f. by that factor. Fitting a single gain k_e on the Fig. 6 e.m.f. with the specified circuit gives $k_e = 1.84$ for 3.46 mH (rms 0.24 kA, peak 11.05 kA at 139 ms, -3.23 kA at 80 ms, 4.33 kA at 300 ms, $I^2 t$ 14.6 MA²s against 14.6) and $k_e = 1.79$ for 3.30 mH (rms 0.31 kA); leaving R free as well returns $R = 23.7$ m Ω , i.e. the physical coil resistance. The whole curve is matched, including the pre-quench dip that depends only on the slow phase

where no eddy effect can act (Fig. 5, left). The conclusion is that Fig. 7 was computed with the circuit of §8.2 and with a disruption whose flux swing is 1.84 times that of Fig. 6 – consistent with Fig. 6 being the worst case for the *voltage* (fast quench) and Fig. 7 the worst case for the *current* (largest flux swing, plasma closest to DIV3), taken from different cases of the ENEA disruption database. Whether the current-worst case also has a larger voltage peak cannot be known from the specification; with $k_e = 1.84$ applied to the Fig. 6 shape the model gives 5.3 kV open-circuit for DIV3, which is the conservative assumption for the converter until ENEA provides the e.m.f. of that case (Sec. 6).

For the total current, Fig. 7 adds a constant 5 kA to the induced current (the difference between the two ENEA curves is 5.0 kA at every instant): the pre-disruption current is not let decay through the crowbar. In the model this corresponds to superposing 5 kA to the result (peak 16.05 kA, I^2t 37.9 MA²s over 300 ms, as ENEA); the physical alternative, $i_0 = 5$ kA decaying with the circuit time constant, gives 12.9 kA and 20.5 MA²s (Fig. 5, right). The ENEA superposition is the contractual one and is conservative.

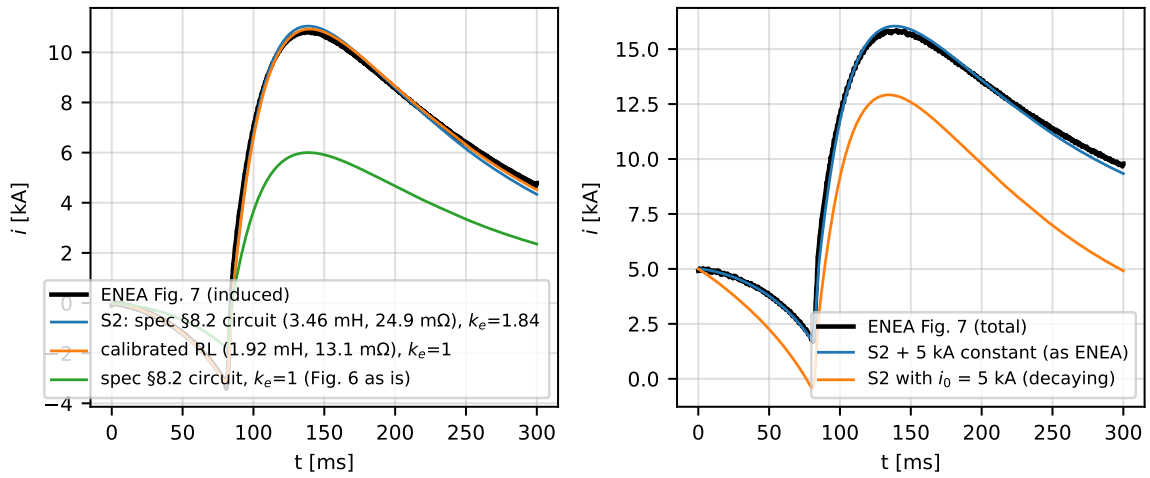


Figure 5: Reproduction of [1] Fig. 7 with an ideal crowbar. Left: induced current with the specified circuit (3.46 mH, 24.9 mΩ) and the Fig. 6 e.m.f. scaled by $k_e = 1.84$ (“S2”, orange), with the black-box R-L of [2] (blue) and with the specified circuit and the Fig. 6 e.m.f. as it is (green). Right: total current with the constant 5 kA superposed as in the specification and with a decaying initial current.

4.2 Effect of the elements neglected in Fig. 7

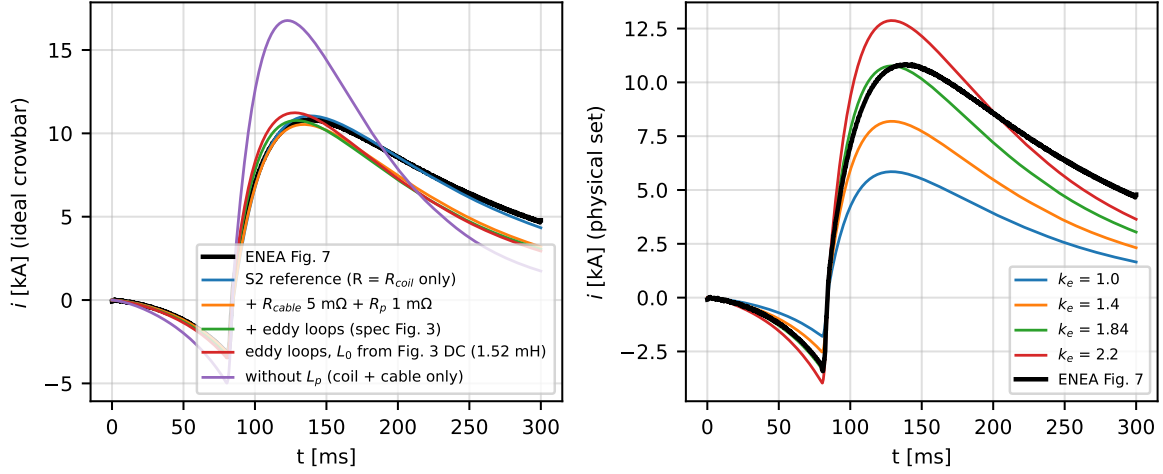
Starting from the ENEA-equivalent setting (S2) the elements that the specification declares neglected were added one at a time (Table 4, Fig. 6): the cable resistance (5 mΩ, worst case of Table 2) and a placeholder 1 mΩ for the protection coil, the two eddy loops of Table 2, and the DC inductance of Fig. 3 instead of the stand-alone value. The resistances reduce the peak by 5% and the I^2t by 18%, because they shorten the tail (3.2 instead of 4.3 kA at 300 ms); the eddy loops raise the peak by 2% (lower inductance on the 25 ms scale) and shorten the tail slightly; the lower DC inductance of Fig. 3 raises the peak by 4%. The spread between the variants is within the accuracy of the reference curves, so the S2 setting is an adequate and conservative representation of the current worst case for the crowbar, while the variants bracket the real circuit. The last row shows why the protection coil exists: without L_p the peak is 16.8 kA (21.8 kA with the 5 kA) and the I^2t is 52% higher. The right panel of Fig. 6 gives the sensitivity to the e.m.f. gain for the fully physical set: the current scales linearly with k_e , from 5.9 kA at $k_e = 1$ to 12.9 kA at $k_e = 2.2$.

4.3 Quench duration and crowbar closing time

Two what-if cases with the analytic e.m.f. illustrate the “current generator” behaviour of the load (Fig. 7). Varying the current-quench duration from 1.5 to 12 ms at constant flux changes the peak

Table 4: Ideal-crowbar current of DIV3 with the Fig. 6 e.m.f. scaled by $k_e = 1.84$, for the ENEA-equivalent circuit and its physical variants. I^2t over 0–300 ms; “total” = induced + 5 kA constant.

Circuit	Peak [kA]	at [ms]	$i(300\text{ ms})$ [kA]	I^2t [MA ² s]	I^2t total [MA ² s]
ENEA Fig. 7 (digitised)	10.9	139	4.79	14.6	38.1
S2: 3.46 mH, $R = R_{coil}$	11.05	139	4.33	14.6	37.9
+ R_{cab} 5 mΩ + R_p 1 mΩ	10.53	134	3.19	12.0	33.5
+ eddy loops (Fig. 3)	10.77	129	3.05	12.2	33.7
+ eddy loops, $L_0 = L_{dc} - L_p = 1.52$ mH	11.24	128	2.94	12.9	34.7
Without L_p (coil + cable, $R_{coil} + R_{cab}$)	16.77	123	1.74	22.1	46.8

**Figure 6:** Left: the physical variants of Table 4. Right: sensitivity of the fully physical set (resistances and eddy loops included) to the e.m.f. gain k_e .

voltage from -6.5 to -2.5 kV ($k_e = 1.84$, DIV3) but the peak crowbar current only from 11.15 to 11.23 kA: the current is set by the flux swing, by the circuit inductance and by the ratio between the delivery time and the circuit time constant, not by the voltage. The converter sees the voltage; the crowbar sees the flux.

The second case delays the closing of an otherwise open circuit (the coil carries no current until the crowbar closes – an idealisation, since the converter freewheeling diodes would conduct). Closing at 85 ms, 2 ms after the spike, gives the same peak as closing at $t = 0$ (10.99 vs 11.05 kA), because 92% of the flux is still to come through the vessel; closing at 100 ms gives 6.2 kA, at 130 ms 2.0 kA. Two consequences for the protection strategy: the crowbar trigger delay (tens of microseconds, [2]) is irrelevant for the current duty, and whatever absorbs flux before the crowbar closes reduces the crowbar current one-to-one. An H-bridge in protection mode applying 600 V against the e.m.f. absorbs about 30 Wb in 50 ms, i.e. roughly half of the 68 Wb of the current worst case, but it must return the corresponding energy (of the order of 300 kJ) to the DC link, which the 665 mF bank cannot take without the fast dump; this is precisely the kind of trade-off the complete Simscape model is meant to quantify.

5 Simscape implementation

5.1 Components

The package `ssc/+dt` contains five components in Simscape language (Table 5); `ssc_build dt` run in the `ssc` folder generates the library `dt.lib`. All electrical ports are Foundation-library electrical conserving ports; the e.m.f. and the plasma quantities are physical signals so that they can come from Simulink (From Workspace, Signal Builder, a lookup table or a plasma model) through a Simulink–PS converter, or directly from the delivered sources.

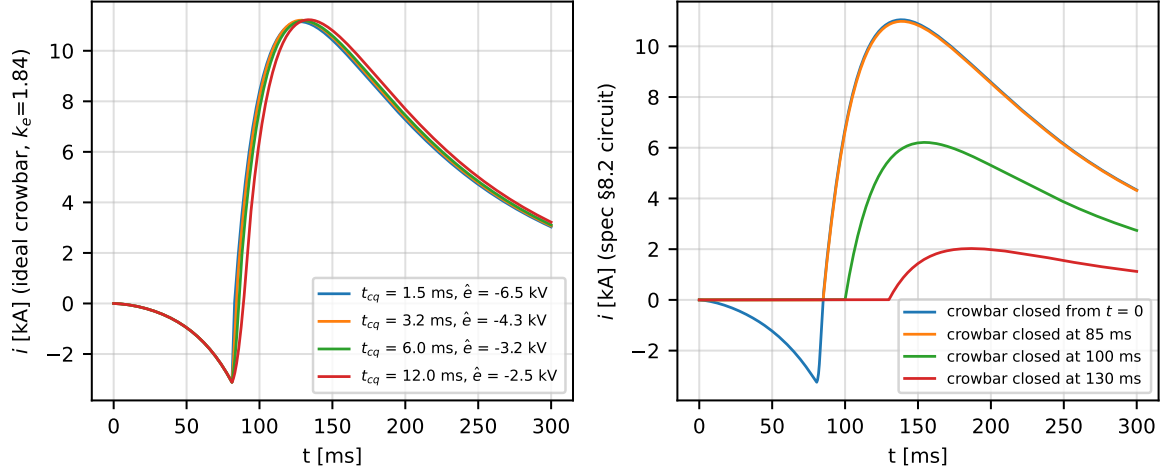


Figure 7: Left: ideal-crowbar current for different current-quench durations at constant flux (analytic e.m.f., $k_e = 1.84$, physical set); the legend gives the corresponding e.m.f. peak. Right: effect of the crowbar closing instant with the coil open until then (S2 setting).

Table 5: Simscape components delivered (package `dtb`, library `dtb.lib`).

Block (file)	Ports	Parameters (defaults)
DIV Coil (<code>div_coil.ssc</code>)	p, n electrical; e PS input [V]; i PS output [A]	R_0 24.9 m Ω , L_0 1.7 mH, L_{m1} 0.593 mH, τ_1 11.8 ms, L_{m2} 0.152 mH, τ_2 102 ms, k_e 1, i_0 0 (DIV3 values; set $L_{m1} = L_{m2} = 0$ for a pure R-L)
DIV Coils 3 (<code>div_coils3.ssc</code>)	p_j, n_j ($j = 1, 2, 3$); e1..e3 PS inputs	$\mathbf{R}_0$ [10.6 18.7 24.9] m Ω , $\mathbf{L}$ diag(0.5, 1.2, 1.7) mH with $M = 0$, $\mathbf{L}_m$, $\boldsymbol{\tau}$ (3×2 , Table 2), $\mathbf{k}_e$, i_{10} , i_{20} , i_{30}
EMF Table (<code>emf_table.ssc</code>)	e PS output	$\mathbf{t_tab}$, $\mathbf{e_tab}$ (vectors, e.g. <code>C.t</code> , <code>C.e_div3</code>), t_0 , k ; linear interpolation, end values held
EMF Analytic (<code>emf_analytic.ssc</code>)	e , Psi PS outputs	Ψ_a , $\Delta\Psi_v$, g , t_q , t_{cq} , τ_v , γ (Table 3, DIV3), t_0 , k
Plasma EMF (<code>plasma_emf.ssc</code>)	Ip [A], Mcp [H] PS inputs; e , Psi PS outputs	τ_v , γ , derivative filter τ_f 50 μ s, k , Ψ_{init} (initial $M_{cp}I_p$)

The coil component declares the currents i , x_1 , x_2 as high-priority initial conditions $(i_0, 0, 0)$, which is consistent with a DC pre-disruption current; the eddy-loop equations contain di/dt and the coil equation dx_k/dt , an index-1 system that Simscape handles like coupled inductors. The Plasma EMF block obtains the flux derivative with a first-order filter ($\tau_f \ll t_{cq}$) and initialises the filter states at Ψ_{init} to avoid a start-up spike; alternatively the “Start simulation from steady state” option of the Solver Configuration block can be used. The analytic source uses the simulation time in conditional equations (`if time - t0 <= tq ...`), which is legal in Simscape language; the piecewise definition creates zero crossings at t_0 , $t_0 + t_q$ and $t_0 + t_q + t_{cq}$.

5.2 Connection and conventions

Fig. 8 shows where the block sits: the cable and the protection coil are external series elements between the crowbar node and the coil; the filter capacitor of the power supply is across the converter side. The sign convention is that of Sec. 2: p to the positive converter terminal, e as plotted in the specification, $i_0 = +5$ kA for the worst case, $k_e < 0$ for the opposite polarity. The block is inductive at its terminals, so it can be connected to any node of the converter model, including a capacitor node, without topological conflicts; a current sensor or the PS output i gives the coil current. When a pre-disruption current $i_0 \neq 0$ is set in the block, the same initial current must be given to the external series inductors (cable, L_p), otherwise Simscape reports conflicting high-priority initial targets.

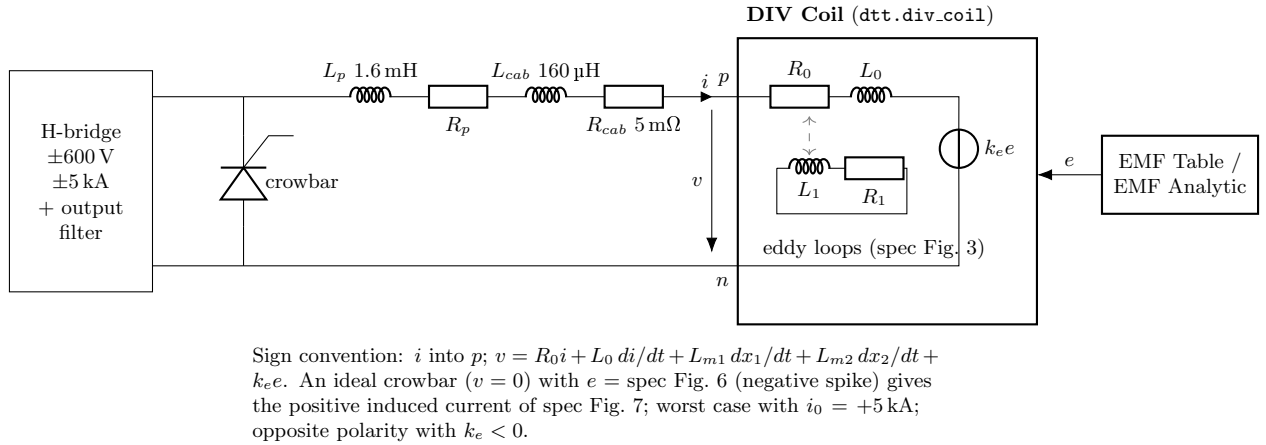


Figure 8: Position of the load model in the DIV3 circuit and sign convention. The validation circuit of Sec. 5.4 is this one with the crowbar closed and the converter removed.

5.3 Parameter sets

The function `dt.div_params.m` returns all the parameter sets of Table 6. The recommended use is: physical set for everything that involves the converter (control, normal operation, overvoltage, protection mode), with the e.m.f. table at $k_e = 1$ for the voltage worst case and at $k_e = 1.84$ for the current worst case; ENEA-equivalent set when the purpose is to reproduce the contractual Fig. 7 (e.g. crowbar $I^2 t$ verification in the FDR); black-box set only as a cross-check.

5.4 Validation procedure

No MATLAB installation was available where this note was written, so the components were written against the Simscape language specification [15] and verified with a Python replica of Eqs. (2)–(3) (exact discretisation of the linear state-space model, `py/divmodel.py`); all figures and numbers of this note come from that replica. Two MATLAB scripts allow the user to close the loop:

Table 6: Parameter sets for DIV3 (`dtc.div_params.m`). Cable and L_p are always external.

Set	DIV Coil block	External / e.m.f.
Physical (<code>par.div3</code>)	R_0 24.9 m Ω , L_0 1.7 mH (or 1.52 mH = Fig. 3 DC $-L_p$), eddy loops of Table 2	cable 160 μ H/5 m Ω , L_p 1.6 mH with its real R_p ; $\mathbf{e}$ = Fig. 6 table, $k = 1$ (voltage) or 1.84 (current)
ENEA-equivalent (<code>par.enea</code>)	R_0 24.9 m Ω , L_0 1.7 mH, $L_m = 0$	cable 160 μ H/0, L_p 1.6 mH/0; $\mathbf{e}$ = Fig. 6 table, $k = 1.84$; reproduces Fig. 7 (rms 0.24 kA); total = + 5 kA
Black box (<code>par.calib</code>)	R_0 13.1 m Ω , L_0 1.92 mH, $L_m = 0$, whole loop lumped	no cable, no L_p ; $\mathbf{e}$ = Fig. 6 table, $k = 1$; cross-check only
Analytic plasma (<code>par.plasma</code>)	physical set	EMF Analytic or Plasma EMF with Table 3; scale k , vary t_{cq} , $\Delta\Psi_v$, t_0

- (1) `dtc.validate_ode.m` (no Simulink needed): loads the CSV, runs the same equations with a trapezoidal integrator (`dtc.sim_coil.m`) and must give the numbers of Table 4 first three rows and of Table 3; this checks the data files and the parameter sets.
- (2) `dtc.build_validation_model.m`: after running `ssc_build dtc` it builds, programmatically, the Simulink model of Fig. 8 (named `dtc.div_validation`) with the crowbar closed (EMF Table, DIV Coil, L_{cab} , L_p and a current sensor in one closed loop), runs it with `ode23t` and overlays the result on Fig. 7; expected: peak 11.05 kA at 139 ms, 4.33 kA at 300 ms. The automatic wiring relies on the port numbering of the generated blocks (physical-signal ports first, as in the Foundation library: left $\mathbf{e}$, p ; right $\mathbf{i}$, n); if it fails the blocks are left in place for manual wiring.

Solver hints: the quench spike has a 3 ms width and the tail 25 ms, so a maximum step of 50 μ s is adequate for the load alone; in the complete model the converter switching (4 kHz, 200 ns jitter) dictates the step. `ode23t` or `ode15s` with relative tolerance 1×10^{-5} ; with the table source the interpolation is linear, so a variable-step solver will not miss the spike.

5.5 Scenarios to explore with the complete model

The following list is the proposed test matrix for the complete power-supply model, in the order in which the results build on each other:

- S1. Ideal crowbar closed from $t = 0$, ENEA-equivalent set: reproduces Fig. 7 (regression test of the model).
- S2. Converter in current control at ± 5 kA and at zero, no crowbar action, $k_e = 1$ and 1.84, both polarities: terminal voltage, current excursion, DC-link voltage, IGBT current – this defines the overvoltage and the current the converter must survive until the crowbar closes and quantifies §8.4 “the induced voltage is high, but it is not totally seen by the PS”.
- S3. Crowbar (real model: diode bridge, thyristor, L_c [3], BOD level) triggered by the terminal voltage or current thresholds: firing instant, thyristor peak, I^2t , energy; then with the firing delayed by 1, 2, 5, 10, 20 ms (the 25 ms flux delivery makes the duty insensitive to small delays, Sec. 4.3).
- S4. Protection mode of the converter (drive the current to zero with ± 600 V) with and without the crowbar, with the fast dump enabled: flux absorbed, DC-link excursion, dump energy – the feasibility limit of managing the disruption without crowbar (§8.4 last paragraph).
- S5. Quench duration 1.5–12 ms, flux $\pm 30\%$ (analytic source): margins on the voltage-based trigger and on the thyristor rating.
- S6. Three coils with DIV Coils 3, all crowbars closing, $M = 0$ and then the ENEA matrix when available; DIV1/DIV2 with their own e.m.f. (-5.9 kV, -60.6 Wb): the current-worst case of the other two crowbars, which the specification does not give.
- S7. Disruption during a 10 Hz full-current waveform (the coil current is not DC when the disruption comes): initial conditions of the eddy loops and of the filter.

6 Limits and open points

- **Fig. 6 and Fig. 7 are different cases.** The gain $k_e = 1.84$ is inferred, not given. ENEA should be asked for the open-circuit e.m.f. of the three coils in the disruption case used for Fig. 7 (and its voltage peak), and to confirm the circuit parameters used for Fig. 7 (coil + cable + L_p , R_{coil} only). Until then the model uses the Fig. 6 shape at $k_e = 1.84$ for the current and, conservatively, also for the voltage (5.3 kV open circuit).
- **Plasma response before the quench.** During the VDE the plasma responds to the coil current; the Thevenin model ignores it (Eq. (1) is exact only after the quench). The effect is confined to the 80 ms of small e.m.f. and to the initial dip of the current, and the fit of the pre-quench dip of Fig. 7 suggests it is small for the shorted circuit; it may matter for the converter in current control, where the coil field acts on the plasma position. ENEA’s “more detailed models” (§4.2) would be the plasma-response linear model (CREATE-L type), which can be inserted as a Simulink block feeding Plasma EMF.
- **Eddy loops above 10 Hz.** The extrapolation of Table 2 beyond 10 Hz is a hypothesis; it affects the voltage division during the 3 ms spike and the ripple current at 4 kHz (the passive structures reduce the inductance seen by the switching frequency), not the crowbar current. The impedance at 100–4000 Hz should be requested from ENEA or measured on the coils.
- **Mutual inductances** between the coils and the e.m.f. of the three coils for the same event are needed for the coupled study S6.
- **Pre-disruption current.** The specification superposes a constant 5 kA; the model can also let it decay. The constant superposition is the contractual duty (15.9 kA, 38 MA² s over 300 ms, 44 MA² s with the tail [2]).
- **Digitisation.** The curves have a 1% full-scale uncertainty; the spike peak is resolved within a few percent. The original ENEA data (time series) should be requested and dropped into the CSV with the same column names.
- **Not compiled.** The .ssc files follow the language specification but were not compiled; ssc_build may ask for cosmetic changes (a unit string, an assert message). The equations are checked by the Python and MATLAB replicas.

7 Files delivered

```
ssc/+dttdiv_coil.ssc div_coils3.ssc emf_table.ssc emf_analytic.ssc plasma_emf.ssc
data/dtt_fig6_fig7_digitised.csv      t_s, e_div12_V, e_div3_V, i_ind_fig7_A, i_tot_fig7_A
data/fig3_fits.json analytic_fits.json gain_fits.json results.json
matlab/dtt_div_params.m dtt_load_curves.m dtt_plasma_scenario.m dtt_sim_coil.m
matlab/dtt_validate_ode.m dtt_build_validation_model.m
py/divmodel.py fit_fig3.py identify_analytic.py compare_models.py fit_gain.py validate_models.
py
README.md div_plasma_load_model.tex fig/
```

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